

Fine-Tuning Qwen2.5-VL for Ancient Tamil Inscription

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Abstract

Ancient stone inscriptions written in scripts such as Tamil and Vattezhuthu are an important source of historical and linguistic knowledge, yet they are hard to read: the text is often damaged, unevenly carved, and written in scripts no longer in common use, and very few labeled datasets exist for training modern models. This project explores how vision-language models can help automatically recognize and interpret such inscriptions. Our system uses Qwen2.5-3B to extract text from inscription images. Because real data is scarce, we generated synthetic images by rendering text with the "Aksharamukha" package and compositing it onto stone backgrounds using the "Augraphy" library. Qwen2.5-3B was then fine-tuned with LoRA on this custom dataset, and a Qwen2.5-0.5B model restored word boundaries to produce readable modern Tamil. The system was evaluated using BLEU and ROUGE. Overall, this work investigates how modern NLP techniques can support the digital preservation and interpretation of historical inscriptions.

1 Introduction

Historical Stone inscriptions particularly those written in archaic scripts like Vattezhuthu and Tamil serve as the primary source of socio-political and linguistic evolution of South Asia. These epigraphic artifacts are not merely relics, they are foundational to understanding the transition of languages like Tamil, which is one of the oldest language in the world, from their ancient roots to modern forms. However, archaeologists face challenges such as physical degradation of stone inscriptions due to centuries of erosion and a declining number of paleography experts capable of deciphering these inscriptions (Diao et al., 2025). To tackle these challenges, this research investigates the use of Vision-Language Models (VLMs) for the automatic recognition and translation of ancient inscriptions. Recent work has shown that multimodal VLMs can be applied effectively to cultural-heritage material, for example to generate descriptive metadata for archival photographs (Facklam et al., 2025). Building on this direction, we propose a pipeline based on the Qwen2.5-3B architecture, which is a transformer-based model

capable of processing both visual and textual information. One of the major challenges in this task is the limited availability of labeled datasets for ancient inscriptions. To address this issue, a synthetic dataset generation framework is developed. The inscription text is first rendered using the "aksharamukha" python package, which converts the modern Tamil script into its ancient predecessors, namely Tamil and Vatteluttu. In addition, the Augraphy library is used to simulate realistic stone textures, lighting variations, and surface degradation. This process helps generate a training dataset that closely resembles real-world epigraphic images. The core of our methodology fine-tunes the Qwen model with Parameter-Efficient Fine-Tuning (PEFT), specifically LoRA (Low-Rank Adaptation) (Dettmers et al., 2023), so it can learn the Tamil and Vattezhuthu scripts without the cost of updating all parameters; a Qwen2.5-0.5B model then converts the recognized text into human-readable form. We evaluate the system with BLEU and ROUGE. Overall, this research supports the digital preservation of cultural heritage by connecting ancient inscriptions with modern Natural Language Processing (NLP) techniques.

2 Dataset

Fine-tuning the Qwen2.5-3B and Qwen2.5-0.5B models required a high-quality, domain-specific dataset. As none is publicly available, our first task was to build one: paired data consisting of an inscription image with its inscription text (for the Qwen2.5-3B model) and the corresponding modern Tamil text (for the Qwen2.5-0.5B model).

The foundational Tamil language text data was sourced from open-source dataset called "Project Madurai" which provided us with Tamil language's unicode UTF-8 text format. To ensure the text was suitable for both visual rendering and sequence modeling, the raw continuous text strings were pre-processed and segmented into meaningful and uniform chunks.

Next, we developed a custom typographical pipeline to automate the conversion of the modern Tamil text chunks (which we obtained in the previous step) to their ancient script equivalent. Furthermore, this pipeline was also responsible for

using the "Aksharamukha" python library to convert the chunks to text images. These renderings depict the ancient script as dark glyphs on a plain light background; we refer to them as the *clean* (*black-and-white*) images, and they form the primary training input for the vision model.

Now, our task is to combine the clean text image and blend it with a particular stone background. We used some sample open-source images as background images. We utilized the "Pillow" python library to combine the text image with the stone background and later used the "Augraphy" python library to add realistic degradation such as erosion, lighting variation and surface noise. This yields, for every clean rendering, a parallel *augmented stone* image. The clean black-and-white renderings are used to fine-tune the vision model, whereas these stone-background images are reserved as a harder, more realistic test condition that simulates weathered real-world inscriptions.

The final output of the pipeline produced a dataset of approximately 28,000 paired text–image samples. To match the input format required by the *Qwen2.5-VL* architecture, the ground-truth data was converted into a conversational JSONL structure. This format explicitly links each rendered image with its corresponding modern Tamil transcription.

3 Methodology

To overcome the dual challenges of severe epigraphic degradation and continuous-script morphology, we implemented a decoupled, two-stage cascaded architecture. A monolithic approach—forcing a single model to perform both visual extraction and linguistic Sandhi resolution (*Puṇarcci*: the morphophonological rules by which adjacent Tamil words fuse together and lose their written boundaries)—yields sub-optimal convergence and exceeds our 20GB VRAM hardware constraint. Therefore, we isolated these tasks into specialized Vision and Linguistic experts.

3.1 Stage 1: Visual Extraction Expert

We utilized the *Qwen2.5-VL-3B-Instruct* model (Bai et al., 2025), leveraging its native Multi-modal Rotary Positional Encoding (mRoPE) (Le and Nguyen, 2026)—a scheme that encodes the two-dimensional spatial layout of image patches alongside the text—to process uneven stone surface topography without aspect-ratio distortion. Be-

cause the base tokenizer lacked extinct Tamil and Vattezhuthu characters, we engineered a vocabulary injection protocol: missing tokens were extracted from our custom tokenizer and safely injected into the 3B model by dynamically resizing the GPU embedding matrix.

To train within the 20GB VRAM limit, we applied QLoRA (Dettmers et al., 2023), which fine-tunes a model whose weights are quantized to 4-bit NormalFloat (NF4), sharply reducing memory use. The low-rank LoRA adapters (rank $r = 16$, with scaling factor $\alpha = 32$ controlling the magnitude of the adapter update) targeted the attention heads, while core MLP layers remained frozen to prevent catastrophic forgetting. Crucially, the `embed_tokens` and `lm_head` layers were explicitly unfrozen to learn the new vocabulary. Utilizing the `paged_adamw_8bit` optimizer—a memory-paged 8-bit Adam variant that offloads optimizer states to avoid out-of-memory spikes—together with expandable memory segments, we successfully trained the model on a highly stratified subset of 6,000 synthetic image-text pairs (split 80/20), strictly balancing the Tamil and Vattezhuthu classes.

3.2 Stage 2: Morphological Segmentation Expert

The historically accurate, continuous string output from Stage 1 is passed to *Qwen2.5-0.5B*, an ultra-lightweight causal language model dedicated exclusively to resolving grammatical Sandhi boundaries and inserting modern spacing. To achieve high accuracy with a sub-billion parameter model, we engineered a sliding-window chunking algorithm across our text corpora. By artificially restricting inputs to 15-word micro-prompts, we generated 25,611 highly localized training pairs (1,000 reserved for validation). This constrained context window forces the model’s attention mechanism to focus entirely on local syntax rather than hallucinating overarching semantic narratives.

4 Results & Evaluation

We evaluate both stages with BLEU (reported on a 0–100 scale) and ROUGE (on a 0–1 scale), the standard metrics for comparing generated text against a reference; for both, higher values indicate closer agreement with the ground truth. Because the base *Qwen2.5-VL* tokenizer does not contain the Tamil and Vattezhuthu characters at all (Section 3.1), the

un-adapted model cannot faithfully produce these scripts and is not equipped to perform the task; the scores below therefore measure recognition ability that did not exist before fine-tuning. As we have no external baseline for these extinct scripts, we use the model’s own performance on clean images as the in-distribution reference point against which its performance under realistic degradation is judged.

We evaluate the vision model under two conditions. First, on a held-out test set of *clean black-and-white renderings*—images of the same type the model was trained on, and thus an in-distribution upper bound—it achieves a BLEU score of 95.40 and ROUGE scores of 0.9881 (ROUGE-1), 0.9657 (ROUGE-2) and 0.9838 (ROUGE-L), confirming that the model learned the extinct scripts well. Second, on the harder *augmented stone images*—the same held-out text rendered onto stone backgrounds with simulated degradation, an out-of-distribution condition that mimics real inscriptions—it achieves a BLEU score of 63.02 and ROUGE scores of 0.8796 (ROUGE-1), 0.7087 (ROUGE-2) and 0.8032 (ROUGE-L). The gap between these two conditions quantifies the cost of realistic visual noise, while the still-high stone ROUGE-1 of 0.8796 shows that character-level recognition is largely preserved. Finally, the segmentation model (Qwen2.5-0.5B) achieves a BLEU score of 72.02 and ROUGE scores of 0.8471 (ROUGE-1), 0.7636 (ROUGE-2) and 0.8469 (ROUGE-L), indicating that it reliably restores word boundaries in the continuous string produced by the vision model and converts it into human-readable modern Tamil.

5 Analysis and Discussion

These results demonstrate that our decoupled approach successfully addresses the epigraphic bottleneck. The drop in the Vision model’s BLEU score from 95.40 on clean images to 63.02 on augmented stone images highlights the severe impact of visual noise (erosion, lighting) on OCR tasks, confirming the physical degradation challenges identified by [Diao et al. \(2025\)](#). However, the preserved ROUGE-1 score of 0.8796 indicates that the model retains strong character-level recognition despite this severe topographical interference. Furthermore, the Segmentation model’s BLEU score of 72.02 proves that a localized, ultra-lightweight language model (0.5B parameters) can efficiently resolve complex continuous-script morphology. Ul-

timately, this validates our core hypothesis: dividing multimodal tasks into specialized visual and linguistic experts, combined with parameter-efficient methods as outlined by [Dettmers et al. \(2023\)](#), is far more efficient under strict hardware constraints than relying on a single monolithic architecture.

6 Future work

Although our system shows promising results for recognizing ancient Tamil scripts such as Vattezhuthu and Tamil, several improvements remain. Future work includes training with larger datasets, exploring a unified multimodal model for direct image-to-text transcription, extending the pipeline to other historical scripts such as Grantha, Brahmi, and early Kannada, and evaluating larger vision-language models such as Qwen3-VL.

7 Examine my experience

Building this cascaded pipeline required a direct, critical application of several course concepts—tokenization, sequence modelling, and transformer attention. Translating these from lab exercises into a working system under strict hardware constraints shifted my understanding of NLP from theoretical modelling to practical, resource-constrained deployment.

7.1 Practical Tokenization: Managing Vocabulary and VRAM Footprints

During the initial labs I viewed tokenization as a routine preprocessing step that chunks raw text into sub-words before feeding a network. That changed in this project: when our base Qwen2.5-VL model failed to recognize the extinct Tamil and Vattezhuthu characters, tokenization ceased to be a preprocessing detail and became a critical architectural bottleneck.

Applying the underlying concepts of sub-word modelling from the labs, I engineered a custom Byte-Pair Encoding (BPE) tokenizer on our cleaned Tamil corpus to extract 5,000 optimal sub-word units. Programmatically injecting these tokens into the 3B model and dynamically resizing the GPU embedding matrix shifted my perspective: I learned that tokenization dictates the physical shape of the model’s memory footprint and its sequence-length efficiency, turning it from a string-splitting utility into an engineering lever for managing GPU memory.

7.2 Hardware-Constrained Fine-Tuning: Adapting LoRA for Vocabulary Injection

The course gave a solid foundation in Transformer internals—we converted a causal GPT model into a bidirectional BERT model—and a hands-on lab on Parameter-Efficient Fine-Tuning (PEFT). Applying LoRA to a small GPT-2 was straightforward, but scaling it to a 3B Vision-Language model required going well beyond that foundation.

To survive our strict 20GB VRAM cluster limit while injecting a new vocabulary, I had to synthesize external scientific literature, specifically the QLoRA method of [Dettmers et al. \(2023\)](#), which shows that fine-tuning over 4-bit quantized weights with low-rank adapters can match full-precision adaptation while drastically reducing memory. I realized that to successfully teach the model a new alphabet, I could not just apply the standard LoRA configuration we used on GPT-2; I had to utilize 4-bit quantization, freeze the core MLP layers to prevent catastrophic forgetting, and explicitly unfreeze the embed_tokens and lm_head layers.

7.3 Mitigating Attention Dilution via Sliding-Window Segmentation

The project also deepened my understanding of causal sequence modelling. Causal models predict each token from preceding context; in ancient Tamil, where words fuse via Pūṇarcci rules without spacing, the model must decide from local cues whether the next character begins a new word. I found that feeding long, unspaced strings into our 0.5B causal model diluted its attention over too long a history, causing severe hallucinations and missed boundaries.

To solve this, I engineered a sliding-window chunking algorithm, restricting the causal sequence to 15-word micro-prompts. By artificially restricting the context history, I forced the model's causal attention to concentrate entirely on local morphophonological syntax rather than distant, irrelevant text. This cemented my understanding that a smaller causal model's attention must be aggressively managed through data structure.

8 Articulate my learning

My most significant learning is that effective NLP system design is not about selecting the largest foundation model but about orchestrating an optimal balance between model architecture, data structure, and hardware constraints. I learned this when

our initial monolithic approach failed: resolving both epigraphic visual noise and linguistic Pūṇarcci rules in one model caused sub-optimal convergence and memory bottlenecks. Decoupling into a 3B Vision Expert and a 0.5B Linguistic Expert, and managing attention dilution via sliding-window chunking, taught me to manipulate sequence modelling and context windows to fit within a rigid 20GB VRAM boundary.

This iterative, problem-solving process directly helped me achieve the course's main learning objective: to seek, assess, and use scientific information within the area of NLP. When confronted with the risk of catastrophic forgetting during our vocabulary injection phase, standard course materials were no longer sufficient. I had to actively seek out specialized literature, assess methodologies like QLoRA, and explicitly use that scientific information to modify our training scripts specifically by unfreezing the embed_tokens and lm_head layers while applying 4-bit NF4 quantization. This challenge successfully transitioned my relationship with academic literature from passive comprehension to active engineering application.

More broadly, this project offers a blueprint for researchers working on low-resource languages and digital heritage preservation, complementing the survey of computational methods for ancient scripts by [Diao et al. \(2025\)](#) and the use of multimodal models on cultural-heritage material by [Facklam et al. \(2025\)](#). Its core takeaway is that state-of-the-art multimodal extraction does not require massive compute: by showing that continuous-script segmentation and high-noise OCR are achievable on academic hardware through architectural decoupling, custom BPE tokenization, and memory-efficient adaptation in the spirit of [Dettmers et al. \(2023\)](#), the methodology democratizes access to epigraphic NLP and points toward deciphering other extinct scripts such as Grantha or Brahmi.

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